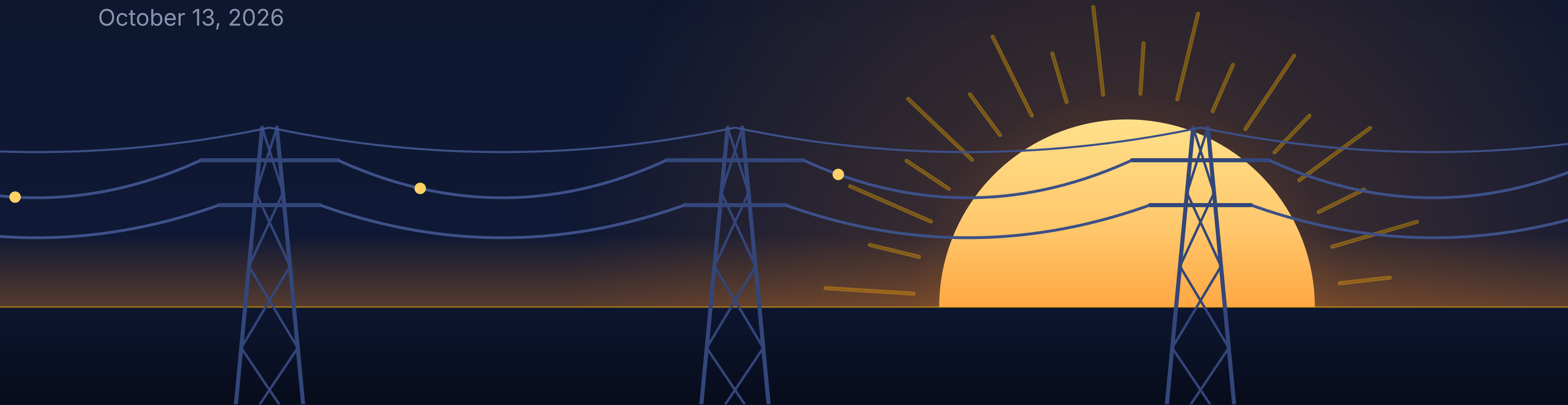
BUS 9430 · CAPSTONE 1 · ISSUE SELECTION AND PLAN

Tomorrow’s Sun, Today’s Plan

Forecasting California's solar power one day ahead

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23,000 MW

of solar power on California's main grid.

All of it depends on tomorrow's weather.

And the plan for tomorrow is locked in today.

Source: CAISO Key Statistics, May 2026 (about 23,000 MW of installed solar as of June 11, 2026).

THE BUSINESS PROBLEM

The plan is locked the day before

TODAY, MORNING

Bids close

Power sellers and buyers tell the market what they can supply and what they need.

TODAY, AFTERNOON

CAISO locks the plan

Which power plants run tomorrow, and how much each one makes.

TOMORROW

The sun shows up. Or it doesn't.

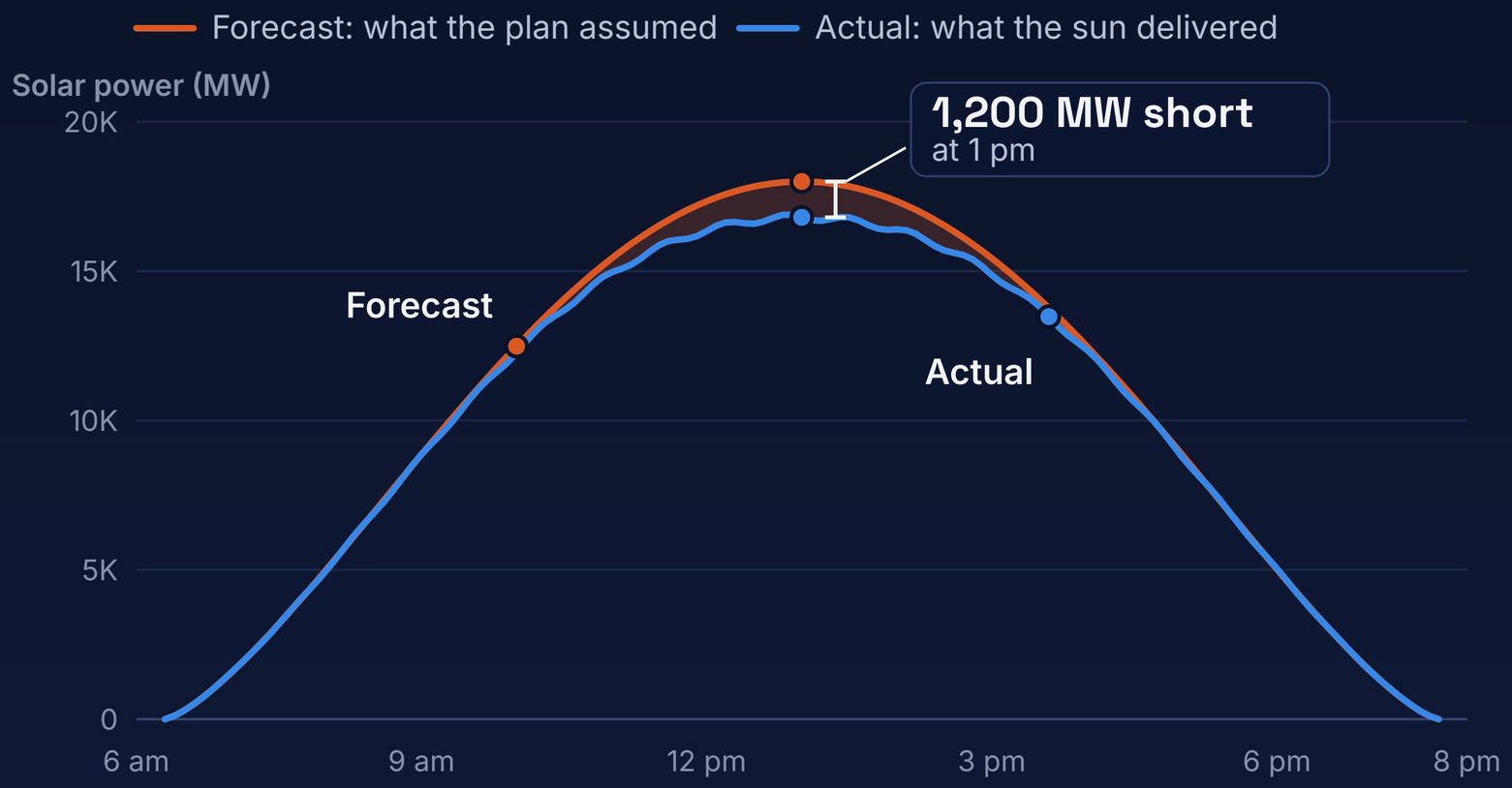
If the solar forecast behind the plan was wrong, operators fix it on the fly.

~80% of California's electricity demand is run by CAISO, serving 32 million consumers. It runs the grid and the market. It owns no power plants.

Source: CAISO Key Statistics, May 2026; California Energy Commission.

WHAT A MISS LOOKS LIKE ILLUSTRATIVE

One cloudy afternoon, 1,200 MW short



FORECAST TOO HIGH

Backup bought at the last minute

Operators counted on power that never arrived.

FORECAST TOO LOW

Gas plants running for nothing

Extra capacity is kept on that the day did not need.

Chart shape is illustrative, not real data. For scale: CAISO's own day-ahead solar forecast is off by about 1,000 MW on average (mean absolute error of 1,082 MW in May and 930 MW in June 2026, as reported by Enverus).

STAKEHOLDERS

Who feels a forecast miss?

Same forecast, four winners.



Grid operators

Fewer last-minute shortfalls.



Utilities and ratepayers

Less spent on emergency power.



Solar and battery owners

Sharper bids and schedules.



Communities and the climate

Less gas burned as insurance.

HOW WE WILL SOLVE IT

Forecast tomorrow’s sun, then race four forecasts

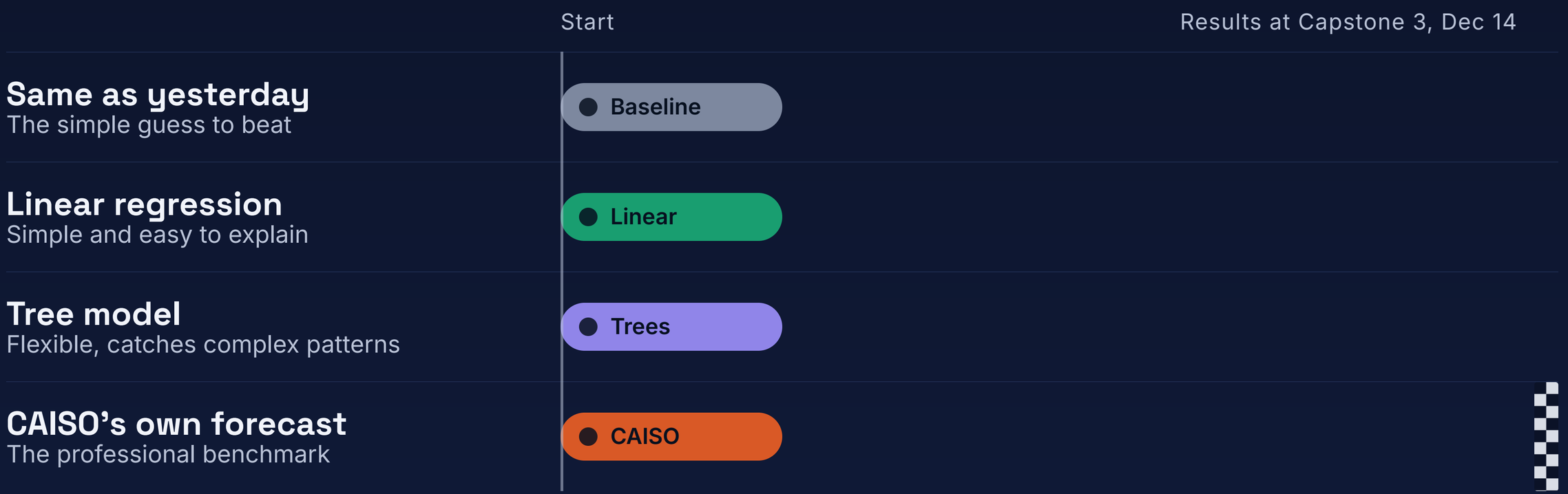
INPUT
Weather forecast from the day before

→

OUR MODEL
Predicts tomorrow’s solar, hour by hour

→

OUTPUT
Megawatts for every hour of tomorrow



THE DATA

Three public datasets, three jobs

THE INPUTS

Weather forecasts from the day before

Cloud cover, sunlight, temperature: what we knew when the plan was locked.

Open-Meteo

THE ANSWER KEY

Solar power actually produced

Every hour, across the CAISO grid. We grade every forecast against it.

U.S. EIA

THE BENCHMARK

CAISO's own day-ahead forecast

The professional forecast we have to match or beat.

CAISO OASIS

All public. Nothing confidential. Hourly data since January 2024: train on earlier months, grade on later ones.

Weather data by Open-Meteo.com (CC BY 4.0).

SUCCESS MEASURES

How we'll know we won

4 to 5% CAISO's own forecast misses by this much, as a share of peak solar output. That is the benchmark.

AVERAGE MISS

≤ 1.25×

CAISO's average miss. Close to the pros, with a model we can explain.

BEAT "SAME AS YESTERDAY"

30%

lower error than the simple guess.

WORST-CASE OVERSHOOT

20%

smaller than the simple guess. This drives how much backup operators hold.

Targets are starting points. We recheck them once model results are in. Benchmark: Enverus analysis of CAISO's day-ahead solar forecast, May and June 2026.

THE PLAN

Nine weeks, three checkpoints



Dates from our project plan. Quiz dates from the course syllabus.

WHY THIS ONE

Matters.

Cleaner, cheaper grid operations for 32 million people.

Measurable.

A clear benchmark to beat, every hour, on public data.

Doable.

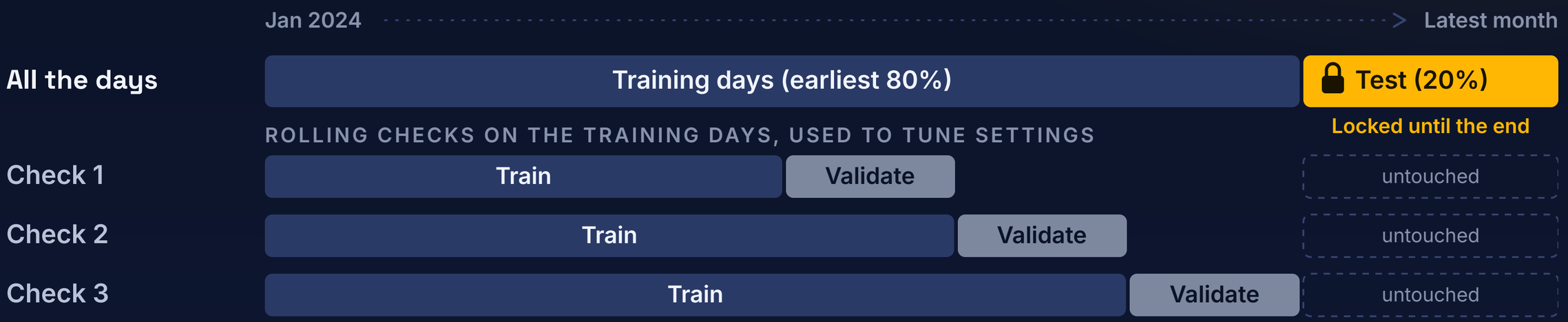
Free data, nine weeks, three checkpoints.

Next: **pull the data** (Oct 14 to 20). Questions?

Appendix below: how we build and test the models, scoring, limits, sources (press the down arrow).

APPENDIX · ANALYTICAL STEPS

Build it, validate it, test it



1

Build

Four models. Inputs are only what we would know the day before.

2

Validate

Rolling checks use earlier days to predict the next block, and pick the settings.

3

Test

The latest 20% of days, scored once, on the same hours as CAISO.

Rolling checks (time-series cross-validation) only ever predict later days from earlier days. The test days are never used for tuning.

APPENDIX · ANALYTICAL STEPS

Comparing models, avoiding overfitting

Comparing the models

- All four forecasts are scored on the **same days and hours**, daylight only.
- The **same scorecard** for each: average miss, improvement over "same as yesterday," worst-case overshoot.
- Compared **day by day**, with an uncertainty range, so we can tell a real gap from luck.
- Results split by **season and weather**, to see when each model struggles.

Avoiding overfitting

- Split by **time, in whole days**. Never at random.
- Tune only on validation. The test days are **scored once**.
- Keep models **modest**: few inputs, limits on how deep and large the trees grow.
- Use only **information known the day before**. No peeking at tomorrow.
- Watch the **gap** between practice and validation scores. A big gap means memorizing.

Overfitting means a model memorizes the past instead of learning the pattern, so it looks great on old days and fails on new ones.

APPENDIX · ANALYTICAL STEPS

Choosing the final model

1

Beat the baseline

At least 30% lower error than "same as yesterday" on validation.

→

2

Win on validation

Best score across all the rolling checks, not one lucky stretch.

→

3

Prefer simple

If two models are close, pick the one we can explain.

→

4

Confirm once

Score the winner on the locked test days against CAISO. Target: within 1.25× CAISO's average miss.

> **The final model is the one that clears every gate. If none does, we report that honestly.**

Targets are starting points and are rechecked once validation results are in.

APPENDIX

Scoring in plain English

MEASURE	IN PLAIN ENGLISH	HOW WE COMPUTE IT	STARTING TARGET
Normalized MAE	On average, how far off are we, as a share of solar capacity?	`mean(abs(forecast - actual)) / installed capacity`, daylight hours only	≤ 1.25× CAISO's
Skill score	How much better than "same as yesterday"?	`1 - model RMSE / baseline RMSE`	≥ 0.30
P95 overshoot	The overshoot we exceed in only 5% of daylight hours. A stand-in for backup needed.	`95th percentile of (forecast - actual)`, in MW	≥ 20% below baseline
MAPE	Percent error, shown only when output is large enough to be meaningful.	`mean(abs(forecast - actual) / actual)`, hours above 5% of capacity	Reported only

APPENDIX

Limits we are watching

Short weather archive. Day-before forecasts are available from January 2024, so each season appears only two or three times.

Curtailment. Reported generation is measured after solar is switched off for oversupply, so "actual" can understate what the sun could produce. We flag those hours.

Utility-scale only. Rooftop solar is mostly missing from hourly grid data.

Time zones. EIA reports in UTC. We convert to Pacific time before matching sources.

Benchmark scaling. Enverus scales CAISO's error by the maximum observed solar output (roughly 23,000 MW). We scale by installed capacity. Close, not identical.

Illustration. The chart on slide 4 shows the idea, not real data.

APPENDIX

Sources

CAISO Key Statistics, May 2026 (installed solar, share of demand, consumers served):
caiso.com/documents/key-statistics-may-2026.pdf

Enverus, day-ahead solar forecast performance across ERCOT and CAISO (CAISO error, May and June 2026):
enverus.com/blog/precision-power-forecasting-day-ahead-solar-forecast-performance-across-ercot-and-caiso

California Energy Commission, who is responsible for transmission:
energy.ca.gov/.../who-responsible-transmission

U.S. EIA, Hourly Electric Grid Monitor and API (solar generation by balancing authority):
eia.gov/opendata

Open-Meteo, Previous Runs API (forecasts issued the day before), CC BY 4.0:
open-meteo.com

CAISO OASIS (day-ahead solar forecast, report SLD_REN_FCST):
oasis.caiso.com

The chart on slide 4 is an illustration, not real data.